Attention Biases in Language Models

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Abstract

Large language models (LLMs) may fail to capture some important information or over-rely on noisy or unimportant parts when understanding lengthy text due to limitations in their attention mechanisms. This can result in unexpected biases in output. Our project aims to mitigate these biases by increasing the attention scores of the critical information and decreasing the scores of noisy or unimportant parts. Specifically, we propose two methods: **data augmentation**, where we replace noisy and unimportant parts with other expressions without changing the original meaning, so that the model can focus on the important part, and **attention intervention**, where we introduce some pre-processing modules to identify important entities and keywords, then increase their importance. Finally, we evaluate the performance of these methods using question-answering datasets.

1 What is the problem

In the process of understanding the text, LLMs may fail to capture critical details or overly rely on unimportant or noisy parts due to their attention mechanisms. This can result in unexpected biases. Our project aims to mitigate these biases.

Why is this problem important: Understanding lengthy background information or user instructions is often the first step for LLMs in solving problems. If biases arise during this step due to the attention mechanism, no matter how powerful the model's performance is, the output may fail to meet expectations. Therefore, addressing this issue is of paramount importance.

What are its key challenges: 1). Spurious features are typically simple and easy to learn, and models tend to memorize these features first. 2). LLMs sometimes generate output based on their prior knowledge, disregarding new factual information from the context, which can lead to incorrect results. 3). Unlike human readers, LLMs struggle to comprehend the emphasis and intent conveyed by the author in the text, such as through bold or italic formatting.

Who would benefit from solving this problem: LLMs always suffer from unanticipated biases in their understanding of the text. Resolving these biases can help LLMs produce more accurate outputs and help many people in different industries. For example, businesses can deliver more precise advertisements to customers, researchers can easily obtain the information they seek, and users of search engines can receive more relevant results.

2 Currently Approaches

2.1 READ Method

The first way is **Robust Natural Language Understanding with Residual Attention Debiasing** [5]. In the paper, they provide a way named **Residual Attention Debiasing (READ)**, which directly mitigates the unintended biases that arise from the language models' attention. Specifically, READ

Preprint. Under review.

method adds a biased attention to help main attention learn residual information, which ultimately results in model paying more attention to informative non-overlapping tokens.

READ method improves out of distribution (OOD) generalization in Natural Language Understanding tasks by explicitly biased attention, thus helping the main model to learn from residual attention. Unlike other methods (e.g. PoE), which apply debiasing on certain level logits, READ is an end-to-end ensemble-based debiasing method, allowing model to learn bias information from bottom to the top. But, the weakness is READ's OOD improvements come at the cost of in-distribution performance, a trade-off seen in most debias methods.

The result shows that, in the experiments, it outperforms other debias techniques across various tasks, including a 12.9% accuracy improvement on the HANS dataset[3] and a 2.7% improvement in F1 score on the PAWS dataset.

2.2 PASTA Method

The second way is **Tell Your Model Where to Attend: Post-Hoc Attention Steering for LLMs**[7]. This approach has a strong connection with our team's proposed approach, which aims to reweight attention scores dynamically for critical tokens. The method of disabling attention heads and reweighting specific tokens provides a precise, task-specific solution.

The method works a post-hoc way, which means it can be applied during inference without any finetune or training. PASTA allows for fine-grained control of the attention mechanism by selecting specific attention heads for reweighting. However, the weakness is, that PASTA's effectiveness relies on task-specific attention profiling. It may require substantial computational resources. Besides, The effectiveness of PASTA largely depends on the user. If the user fails to accurately specify which parts of the information are important, the model may not be able to focus on the key information. We plan to approach this issue by introducing an additional model.

The results show that PASTA improves both instruction-following and contextual comprehension. For example, PASTA boosts the format accuracy by over 22% on JSON formatting tasks with LLAMA-7B.

2.3 Product of Experts Method

The third paper **End-to-End Self-Debiasing Framework for Robust NLU Training** [1] provides a way named **PoE**. The authors propose an end-to-end debiasing framework. This approach doesn't require additional training stages or manually crafted bias features. The bias model identifies lexico-syntactic biases. There is a noise vector applied to the main model's representations to reduce its reliance on these biased features.

The advantage is it allows for simultaneous training of both the bias and main models. This way ensures that the bias model helps locate and mitigate biases during training. The approach is computationally efficient and requires minimal additional parameters. However, The method requires noise addition and example reweighting, which may introduce complexity in managing both models' interactions. Besides, PoE just relies on top of the main model's inter- mediate representations, which are highly compressed and the propagation may suffer from information loss, thus providing limited debiasing signal to low-level attention..

The method achieved 83.2% accuracy on the MNLI development set (in-domain) and 71.2% on the HANS dataset (OOD). The method achieved 90.2% on the QQP development set and 46.5% on PAWS. But without noise addition, the OOD performance decreased across tasks, especially on HANS, dropping from 71.2% to 68.6%. This shows the importance of adding noise to de-emphasize bias features.

3 How to approach it

Prior research [7] and [5] have introduced that when prompts containing substantial background information, LLMs or language models (LMs) often struggle to capture the most critical details. The phenomenon causes the model to be overwhelmed by excessive contextual information, leading to incorrect outputs. Visualizations from PASTA [7] and READ [5] reveal that the content bias primarily results from inaccuracies in the attention distribution. Therefore, the primary challenge is how to direct the attention mechanism to focus on the most relevant aspects, such as reducing attention on noisy elements while emphasizing the important ones. Our team aims to reduce the score of less relevant parts and increase the the score of key terms. We will address this issue from two perspectives: data augmentation and attention intervention.

Data Augmentation: Data augmentation is widely used in industry due to its advantage of not requiring changes to the model's architecture. Our approach is to replace the noisy or unimportant portions of the input with alternative expressions, while maintain essential meaning of the input's key semantics. This helps the model to learn the significance of the important parts more robustly and minimize the influence of irrelevant content. However, identifying the unimportant parts will be a challenge. We may need to use additional models to determine the non-essential sections in a passage. If the accuracy of these models is too low, it could lead to the failure of this method.

Attention Intervention: Firstly, inspired by PASTA [7], we propose an explicit intervention in the attention mechanism by highlighting the key information. This idea comes from human writing habits. In human writing, we often emphasize key information by using bold, quotations, or other markers to signal their importance. Similarly, we plan to utilize other models, such as entity recognition model to extract key entities or a keyword weighting model to identify the most important terms. Then these important terms will be marked with specific symbols (e.g., quotes or brackets). We believe that this human writing habits have been extensively learned by the model during its pre-training stage. Therefore, such input should be able to effectively guide the attention mechanism. Similarly, if the accuracy of these models is too low, it could result in the failure of our method. Secondly, inspired by READ [5], we plan to introduce auxiliary modules to specifically learn attention biases. However, we haven't figure out how to manage this module. We will add details after our further research and analysis.

Finally, we will validate our method on question-answering datasets, such as SQuAD (Stanford Question Answering Dataset)[4] and HotpotQA[6]. These two datasets provide long contexts and questions that closely align with our goal. Due to computational resource limitations, we plan to use GPT-2 [2] as our backbone. We will compare our approach with pretrained language models like GPT-2, Pasta[7], POE[1] and READ[5], evaluating accuracy and recall on these datasets. We will also compare the attention scores from visualizations to assess the effectiveness of our method in guiding attention.

4 Project Timeline

Week 5: 1) Set up the training environment on Google Cloud. 2) Continue reading papers related to attention debias.

Week 6: 1) Conduct attention score visualization on GPT-2 to analyze and validate the feasibility of our ideas. 2) Identify common characteristics of data that do not perform well and examine their specific behavior within the model. 3) Finalize the technical approach for attention intervention.

Week 7: 1) Implement code for data augmentation and validate the effectiveness of the method. 2) Implement code for highlighting important parts of the input and validate the effectiveness of the method. 3) Compare these methods with the baseline performance of the backbone model.

Week 8: Complete the coding for the attention mechanism and begin full-scale experiments.

Weeks 9-10: 1) Adjust the methods and continue experimentation. 2) Conduct comparative analysis with other existing approaches.

Week 11: Perform visualization experiments and ablation studies to analyze the effects and underlying reasons for the observed results.

Week 12: Summarize the methods and outcomes of the project. Prepare for the project presentation.

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